Diet Optimization

Valluri Leela Sai Krishna, Omhari Indoria, Pawan Kumar

Indian Statistical Institute

Introduction

- Unawareness about overall nutrient consumption
- Focus on balancing a person's daily nutritional intake by optimizing their fourth meal of the day
- Dataset:
  - Anuvaad Indian Nutrition Databank
  - Indian Council of Medical Research, MoFHW, Gov of India

Plan & Problem

- Followed PPDAC cycle
- Shannon entropy (motivation): This measures the nutritional diversity or "evenness" of a food item.

$$H = -\sum_{i=1}^{k} p_i log(p_i)$$

  It considers a single distribution.
- Kullback–Leibler Divergence: This measures how much a specific food's nutrient profile (p) diverges from a target profile (r= RDA - Consumed; in proportions)

$$D_{KL}(p \parallel r) = \sum_{i=1}^{k} p_i \log \left(\frac{p_i}{r_i} \right)$$

- p_i and r_i satisfy the three Kolgomorov's axioms of probability
- Returns the optimum nutrient proportions for the fourth meal given the consumed meals

Data

- Anuvaad Indian Nutrition Databank: Contains 1014 food items (mostly Indian, from from the US and the UK); 40 nutrients (per serving, per 100 g); food_name, primary_source
- Indian Council of Medical Research
  1. Common nutrients: magnesium, iron, zinc, B3 (niacin), B6, B9 (folate), vit C, vit A, vit D
  2. Body type (six out of twenty-three): age group, physical activity, gender
- Unavailability of data for other body types

Analysis

- We have used KL Divergence as our main tool for our Analysis.
- In the KL Divergence,
  - Target Distribution(R): $(r_1, r_2, ..., r_9)$, where $r_i = \frac{t_i}{\sum_9 t_j}$. Here, $t_i = |RDA_i - Consumed_i|$ for i^{th} nutrient.
  - True Distribution(X_j): $(x_{j1}, x_{j2}, ..., x_{j9})$, where x_{ji} = proportion of i_{th} nutrient for j_{th} food item.
- M and N are the TUL (upper bound) and EAR (lower bound) proportion vectors respectively for 9 nutrients.
- F_j represents the j_{th} index food item in the dataset.

Objective Function (optimal food item)

$$\hat{F} = \arg\min_{F_j} \sum_{i=1}^{9} x_{ji} \log\left(\frac{x_{ji}}{r_i}\right)$$

- It helps to identify the optimal food item(from the given dataset) to consume as the fourth meal keeping the divergence from RDA values as small as possible for the day.
- $\hat{F}$ is the Optimal Food Item as the fourth meal to consume.

Objective function (Optimal Nutrient Proportions)

$$\begin{aligned} \text{minimize} \quad & \sum_{i=1}^{9} x_i \log\left(\frac{x_i}{r_i}\right) \\ \text{subject to} \quad & x - M \preceq 0, N - x \preceq 0, \\ & \sum_{i=1}^{9} x_i = 1 \end{aligned}$$

- It finds the optimal nutrient proportions not corresponding to any food item in the dataset.
- Where $x \in \mathbb{R}^9$, and let x^* is the optimal solution for the objective function.
- Then $x^* = (x_1^*, x_2^*, .., x_9^*)$ is the optimal nutrient proportions vector, where x_i^* is the ith optimal nutrient proportion.

Tools

- Exploited features in R like-
  1. "pattern matching and replacement"
  2. "split, apply and combine"
- Additional columns like category (desserts, bread, curry etc.), breakfast, lunch etc. were added using grepl() function
- Prefernce of CVXPY (Python) over CVXR (R); less dependencies e.g., rust, fortran
- Inbuilt function: kl_divergence()

Instances of the Unconstrained Problem


```
> rda_sub[n,]  ##rda values in grams
     age_group age_yrs weight magnesium  iron   zinc vit_b3 vit_b6  vit_b9 vit_c   vit_a   vit_d
6 Adult Women     >=18     55      0.37 0.029 0.0132  0.018 0.0024 0.00022 0.065 0.00084 1.5e-05
> rda_prop[n,4]
[1] "Heavy"
> consumed[,1] ##in proportions
[1] "Soya chunks korma (Nutrinugget korma)"
[2] "Cucumber sharbat (Kheere ka sharbat)"
[3] "Jackfruit kofta curry (Kathal ka kofta curry)"
> x[1:5,] ##bottom 5 objective values
                                          food_name      val
150       Cabbage manchurian (Pattagobhi manchurian) 1.590852
27                                Chinese fried rice 1.593766
79                                       Pasta salad 1.634563
63  Cauliflower kofta curry (Phoolgobhi kofta curry) 1.642327
135                                    Roast potatoes 1.658636
> colSums(b[,2:10])+ (a[1,2:10])  ##final nutrient intake in grams
     magnesium        iron        zinc      vit_b3       vit_b6       vit_b9      vit_c
119 0.06887769 0.002592784 0.001287373 0.001297926 0.0003740753 7.350099e-05 0.02973647
           vit_a        vit_d
119 4.376007e-05 4.521586e-06
```

```
> rda_sub[n,]  ##rda values in grams
     age_group age_yrs weight magnesium  iron   zinc vit_b3 vit_b6  vit_b9 vit_c   vit_a   vit_d
6 Adult Women     >=18     55      0.37 0.029 0.0132  0.018 0.0024 0.00022 0.065 0.00084 1.5e-05
> rda_prop[n,4]
[1] "Heavy"
> consumed[,1] ##in proportions
[1] "Sour lentils (Khatti dal)"      "Coffee drops"
[3] "Rice flakes and roasted channa"
> x[1:5,] ##bottom 5 objective values
                                          food_name       val
27                                Chinese fried rice 0.5904691
150       Cabbage manchurian (Pattagobhi manchurian) 0.5965593
61  Lotus stem kofta curry (Kamal kakdi kofta curry) 0.5975013
79                                       Pasta salad 0.5988983
151                                    Macaroni salad 0.6023429
> colSums(b[,2:10])+ (a[1,2:10])  ##final nutrient intake in grams
    magnesium        iron        zinc      vit_b3       vit_b6       vit_b9      vit_c
119 0.1321252 0.007831993 0.002913844 0.002747504 0.0004199922 7.411918e-05 0.01991214
           vit_a       vit_d
119 0.0002106165 1.21684e-05
```

Instances of the Constrained Problem

Valluri Leela Sai Krishna, Omhari Indoria, Pawan Kumar

```
>>> rda_prop.iloc[body,]
age_group            Adult Women
age_yrs                      >=18
weight                       55.0
physical_activity           Heavy
magnesium                0.741966
iron                     0.058154
zinc                      0.02647
vit_b3                   0.036096
vit_b6                   0.004813
vit_b9                   0.000441
vit_c                    0.130345
vit_a                    0.001684
vit_d                     0.00003
Name: 5, dtype: object
>>> consumed.iloc[:,0]
236                                        Pasta salad
101                              Tandoori parantha/paratha
781    Mango milkshake with ice cream (Aam milkshake ...
Name: food, dtype: str
>>> print("Status:", prob.status)
Status: optimal
>>> if prob.status == 'optimal':
...     print("Optimal Nutrients:\n", var.value)
... sum(var.value)
Optimal Nutrients:
 [4.94271430e-01 3.34070304e-03 1.78774663e-03 3.17694172e-04
 3.45429242e-02 2.61439199e-04 4.65447962e-01 9.10964009e-06
 2.09891644e-05]
np.float64(0.999999998780813)
```

```
>>> rda_prop.iloc[body,]
age_group            Adult Women
age_yrs                      >=18
weight                       55.0
physical_activity           Heavy
magnesium                0.741966
iron                     0.058154
zinc                      0.02647
vit_b3                   0.036096
vit_b6                   0.004813
vit_b9                   0.000441
vit_c                    0.130345
vit_a                    0.001684
vit_d                     0.00003
Name: 5, dtype: object
>>> #rda_prop.iloc[body,3]
>>> consumed.iloc[:,0]
179                     Gravy for kofta
528              Minced meat ball curry
15     Mango milkshake (Aam milkshake)
Name: food, dtype: str
>>> print("Status:", prob.status)
Status: optimal
>>> if prob.status == 'optimal':
...     print("Optimal Nutrients:\n", var.value)
... sum(var.value)
Optimal Nutrients:
 [4.71729849e-01 6.36958801e-03 8.00627796e-03 1.34911728e-02
 3.76136396e-02 3.45352682e-04 4.61205058e-01 1.18129311e-03
 5.77651472e-05]
np.float64(0.9999999964380655)
```

Limitations

- Neglects individual variabilities such as allergies, vegetarianism, etc.
- Nutrient weights (x_i) don't add up linearly. Some minerals stop others from being absorbed.
- KL-Divergence returns undefined(NaN) values when a nutrient's proportion (x_i) is exactly zero.
- The consumed and the suggested optimal nutrients' amount are for 100 g of every item. A square meal weighs 450 g and a dessert less than 50 g.
- The analysis ignores total calorie intake, the recommendations might lead to either calorie deficiency or excess calorie consumption.
- The analysis assumes accurate measurements from the dataset.

Conclusion & Future Scope

- Formulated daily dietary planning as a convex optimization problem & calculated the ideal nutritional profile for the fourth meal.
- Using KL Divergence as objective function, the divergence between
  1. final meal's nutrient proportions (x)
  2. the target RDA (R)

  is minimized (bounded below & above the EAR & TUL)
- Suggests an ideal food item with the appropriate nutrient proportions.
- In the future, adding calorie intake and suitably weighed food can make the model more useful in real life.

Thank You :)